

SANANDAJ, Iran, Islamic Republic of

WMO: 407470

Lat: 35.247N

Lon: 47.009E

Elev: 1378

StdP: 85.83

Time Zone: 3.50 (IRN)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-9.8	-7.2	-14.8	1.2	-0.7	-12.8	1.5	2.1	10.9	8.3	9.4	7.4	0.6	10	0.498

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.4	39.1	16.3	38.0	16.1	36.8	15.8	17.7	34.5	17.1	34.3	16.5	33.8	3.9	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
11.9	10.3	21.3	10.9	9.6	20.3	9.9	9.0	19.2	55.5	34.4	53.2	34.4	51.3	33.8	20.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.5	7.8	6.5	-12.9	41.1	3.3	0.8	-15.3	41.6	-17.2	42.0	-19.1	42.5	-21.5	43.0		
(5)			-13.2	18.6	3.3	1.1	-15.6	19.3	-17.5	20.0	-19.3	20.6	-21.7	21.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	14.5	1.5	3.8	8.5	12.7	17.9	24.1	27.9	27.4	21.8	15.7	8.2	3.8		
	DBStd	9.57	4.14	3.83	3.56	3.15	2.63	2.61	2.16	2.15	2.54	2.87	3.33	3.21		
	HDD10.0	786	265	174	72	11	0	0	0	0	0	1	72	192		
	HDD18.3	2289	523	406	307	169	40	0	0	0	3	88	304	449		
	CDD10.0	2434	0	2	24	93	244	423	556	539	355	179	17	1		
	CDD18.3	896	0	0	1	1	26	174	298	281	107	8	0	0		
	CDH23.3	14569	0	0	9	59	594	2781	4599	4331	1882	314	1	0		
	CDH26.7	8368	0	0	0	11	173	1547	2911	2717	946	64	0	0		
(14) Wind	WSAvg	2.1	2.0	2.2	2.8	2.5	2.2	2.4	2.3	2.2	1.9	1.8	1.6	1.6		
(15) Precipitation	PrecAvg	409	43	47	61	77	29	4	3	1	2	32	61	45		
	PrecMax	652	86	108	179	143	116	35	18	7	8	147	213	110		
	PrecMin	271	4	5	10	26	0	0	0	0	0	0	4	3		
	PrecStd	89	21	22	34	37	26	6	4	1	3	34	44	25		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	14.1	17.1	24.8	28.1	32.1	38.2	40.9	40.1	36.2	30.2	21.9	17.1		
		MCWB	5.5	7.1	9.9	12.4	14.0	15.8	16.8	16.5	15.1	12.8	9.8	6.9		
	2%	DB	12.1	14.8	21.1	24.9	30.2	36.9	39.4	39.0	34.9	28.2	19.2	14.0		
		MCWB	5.0	6.1	8.7	11.8	13.3	15.4	16.4	16.3	14.5	12.1	9.0	6.3		
	5%	DB	10.2	12.8	18.9	22.9	28.8	35.2	38.2	37.9	33.5	26.9	17.2	11.9		
		MCWB	4.2	5.4	7.8	10.7	12.8	14.9	16.2	16.0	14.0	11.7	8.4	5.5		
	10%	DB	8.2	10.8	16.4	20.9	26.9	33.9	37.0	36.8	32.0	24.9	15.1	9.9		
		MCWB	3.6	4.7	6.9	10.1	12.6	14.5	16.1	15.8	13.6	11.2	7.8	4.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.5	8.8	10.7	13.7	16.3	17.9	18.8	18.7	16.4	14.0	11.7	8.7		
		MCDB	11.3	14.3	20.5	23.3	27.0	32.8	35.6	35.9	32.1	25.6	17.4	13.8		
	2%	WB	6.1	7.2	9.6	12.8	14.9	16.4	17.9	17.6	15.4	13.1	10.5	7.4		
		MCDB	10.0	12.6	18.9	21.8	25.4	32.8	34.8	34.8	31.9	24.1	15.5	12.0		
	5%	WB	5.0	6.1	8.6	11.9	14.1	15.7	17.2	17.0	14.7	12.5	9.6	6.3		
		MCDB	9.1	11.2	16.9	19.7	24.6	33.0	34.8	34.2	31.1	23.4	14.5	10.4		
	10%	WB	3.9	5.1	7.7	11.0	13.4	15.1	16.6	16.4	14.1	11.8	8.7	5.3		
		MCDB	7.9	9.8	15.1	18.6	23.9	32.3	34.3	34.0	30.2	22.7	14.0	9.0		
(35) Mean Daily Temperature Range	5% DB	MDBR	10.9	12.0	13.2	13.9	16.2	19.1	18.4	19.3	20.3	17.2	13.0	11.4		
		MCDBR	14.3	15.5	17.7	17.8	19.6	20.4	20.0	20.4	21.5	20.5	17.1	15.0		
	5% WB	MCWBR	8.1	8.0	7.8	7.0	6.4	6.2	6.0	6.1	7.3	8.0	8.2	8.2		
		MCWBR	12.1	12.8	14.8	14.5	16.1	18.7	17.6	18.0	19.9	17.1	12.8	12.5		
(40) Clear-Sky Solar Irradiance	taub		0.325	0.373	0.439	0.529	0.520	0.468	0.507	0.469	0.404	0.428	0.359	0.326		
	taud		2.246	2.099	1.939	1.765	1.811	1.943	1.853	1.937	2.094	2.006	2.200	2.274		
	Ebn at Noon		862	852	824	768	782	822	787	809	848	779	811	835		
	Edh at Noon		105	138	176	221	214	187	204	184	149	149	110	97		
(44) All-Sky Solar Radiation	RadAvg		2.77	3.77	4.80	5.74	6.97	8.31	7.83	7.26	6.21	4.40	3.05	2.44		
	RadStd		0.22	0.34	0.39	0.47	0.47	0.36	0.26	0.24	0.27	0.33	0.37	0.28		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (5 neighbors)	N/A	N/A	N/A	+0.71	-1.56	-0.93	N/A	N/A	N/A	N/A

Nomenclature: See separate page